MTH636 – Game Theory

Lecture 4: Weak Dominance, Auctions & Max-Min Strategies

1 Iterative Elimination of Weakly Dominated Strategies

Recall from the previous lecture that the elimination of strictly dominated strategies one by one is called **Iterative Elimination of Strictly Dominated Strategies (IESDS)**. If this process gives a unique strategy profile, we call it a solution to the game.

We can apply a similar principle for *weakly* dominated strategies.

Key Insight: Assumption 1'

A rational player does not use a **weakly dominated strategy**.

Definition: Rational Outcome

A strategy profile is called **rational** if it is the unique result of an iterative elimination of weakly dominated strategies.

Note

As shown in the previous lecture, the order of elimination matters for weakly dominated strategies. If different orders lead to different unique outcomes, the solution might be ambiguous.

2 Second Price Sealed Bid Auction (Vickrey Auction)

A classic application of weakly dominated strategies is the second-price sealed-bid auction.

2.1 Formal Setup

- **Players:** $N = \{1, 2, \ldots, n\}$
- **Strategies (Bids):** $S_i = [0, \infty)$ for each player i. Let player i's bid be b_i.
- **Valuations:** Each player has a true valuation v_i for the item.
- **Payoff Function:** $u_i(b_1, \ldots, b_n)$ is defined as:

$$u_i(b_1, \ldots, b_n) = \begin{cases} 0 & \text{if } b_i < \max_{j \in N} b_j \\ \dfrac{v_i - \max_{j \neq i} b_j}{|\{k \mid b_k = \max_{j \in N} b_j\}|} & \text{if } b_i = \max_{j \in N} b_j \end{cases}$$

Intuition

How it works: The highest bidder wins the item but pays the *second-highest* bid ($\max_{j \neq i} b_j$). If there is a tie for the highest bid, the tied players split the surplus equally (represented by the denominator).

Note

Since players do not know each other's valuations v_j, this is **not** a complete information game. It is a game of incomplete information.

2.2 Truthful Bidding is a Weakly Dominant Strategy

Theorem: Vickrey's Theorem

In the second-price sealed-bid auction, **bidding your own valuation weakly dominates all other strategies.**

In other words, for player i, playing $b_i = v_i$ is always at least as good as any other bid b_i', regardless of what the other players do:

$$u_i(v_i, b_{-i}) \geq u_i(b_i', b_{-i}) \quad \forall b_{-i} \in [0, \infty)^{n-1},\ \forall b_i' \in [0, \infty)$$

Proof Reference: See Section 4.6, Theorem 4.15 in Solan's Game Theory textbook for the detailed proof. The intuition is that changing your bid from v_i only changes the outcome when it changes whether you win or lose, and in those cases, deviating from v_i either makes you lose when winning was profitable, or makes you win when the price is higher than your valuation.

3 The Max-Min Approach

Another way to approach a game is to think defensively. What is the absolute worst that can happen if I play a certain strategy, and how can I make that worst-case scenario as good as possible?

Example: Defensive Play Example

Consider the following game matrix from Player 1's perspective:

$P_1 \setminus P_2$	L	R	**Row Min**
T	(2, 1)	(2, −20)	**2**
M	(3, 0)	(−10, 1)	-10
B	(−100, 2)	(3, 3)	-100

For each strategy, Player 1 looks at the minimum possible payoff:

- If T is played: min(2, 2) = 2
- If M is played: min(3, -10) = -10
- If B is played: min(-100, 3) = -100

The maximum of these minimums is 2, achieved by playing T. Strategy T guarantees

Player 1 a payoff of at least 2, regardless of Player 2's action.

3.1 Formal Definitions

Suppose player i plays strategy s_i. The minimum amount he/she will get is:

$$\min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$$

Definition: Max-Min Value

The **max-min value** of player i, denoted $\underline{v}_i$, is the highest payoff player i can guarantee for themselves:

$$\underline{v}_i = \max_{s_i \in S_i} \left[\min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) \right]$$

Definition: Max-Min Strategy

Suppose $s_i^* \in \arg\max_{s_i \in S_i} \left[\min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) \right]$.
Then s_i^* is called a **max-min strategy** for player i.

Key Takeaway

By playing a max-min strategy, a player can ensure they receive at least their max-min value, no matter how the opponents play:

$$u_i(s_i^*, s_{-i}) \geq \underline{v}_i \quad \forall s_{-i} \in S_{-i}$$